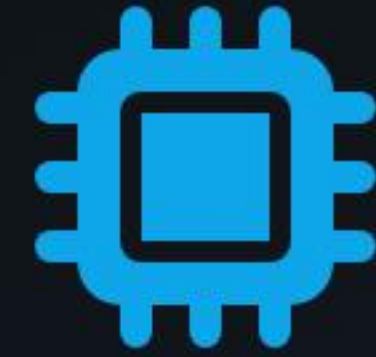




SafeDrive AI

Real-Time Drowsiness Detection using Vectorized Deep Learning

Engineering Analysis & Final Architecture Review

The Fatigue Crisis

Microsleeps are unpredictable, fatal, and occur in dynamic, noisy environments where traditional monitoring fails.

- ⚠️ **The Challenge:** Real-world cabin lighting and rapid head movements corrupt standard camera streams.
- 🧠 **The Solution:** A robust AI pipeline that analyzes *temporal patterns* rather than static single frames.
- 🎯 **Objective:** Achieve high recall (zero fatal misses) with minimal computational latency.

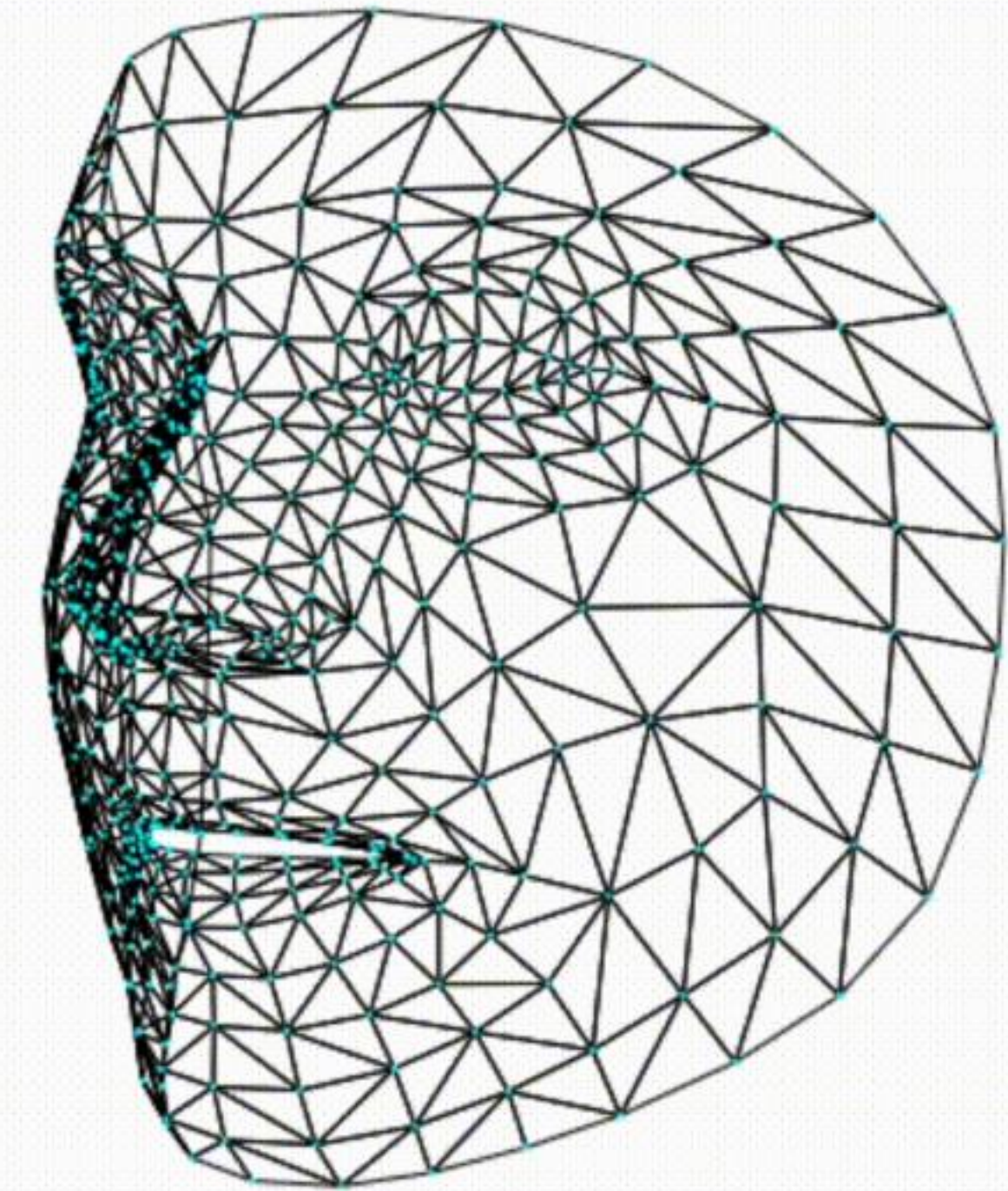


Feature Extraction Engine

MediaPipe Face Mesh

We extract 468 3D facial landmarks to compute scale-invariant biometric ratios.

- 👁️ **Eye Aspect Ratio (EAR):** Dynamically calibrated against a 50-frame personal baseline.
- Mouth Aspect Ratio (MAR):** Captures early signs of fatigue (yawning).
- ↕️ **Head Pitch:** Monitors subtle forward head drops.
- 👁️ **PERCLOS:** The medical gold standard—Percentage of Eye Closure over a moving window.



Architectural Shift: Vectorization

Solving Temporal Redundancy

Sequential networks (LSTMs) struggled with the repetitive nature of 30FPS video, leading to extreme overfitting or underfitting (Recall stuck at 44%).

We engineered a paradigm shift: converting 3D time-series data into 1D Statistical Vectors. By mathematically summarizing a window, we eliminate noise and capture pure behavioral intent.

$$V = \mu_{\text{EAR}} , \sigma_{\text{EAR}} , \min (\text{EAR}) , \dots , \Delta_{\text{Pitch}}$$

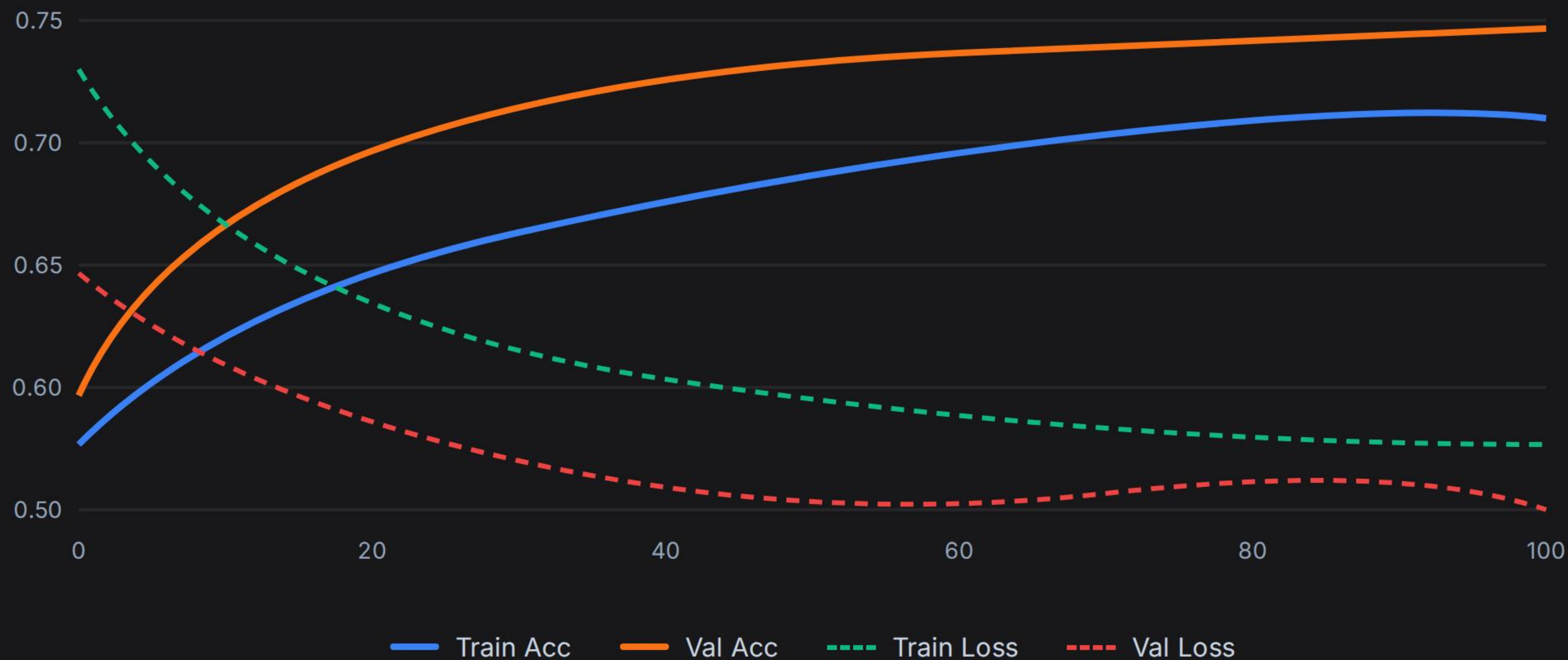
~~60 Frames~~



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STATISTICAL FEATURES

Training History: Stability & Generalization



i Validation Accuracy > Training Accuracy due to aggressive **Dropout (0.4)** during training. The DNN generalizes perfectly without overfitting.

Final Confusion Matrix (Vector Dataset)

		Predicted Label	
True Label	Awake (0)	1688 True Negative	332 False Positive
	Drowsy (1)	757 False Negative	1364 True Positive

By vectorizing the data, **Drowsiness Recall surged to ~64%** on highly noisy, unseen test data. The total sample size (4141) reflects 2-second consolidated windows, proving operational robustness.

Academic Classification Report

CLASS	PRECISION	RECALL	F1-SCORE	SUPPORT
Awake (0)	0.69	0.84	0.76	2020
Drowsy (1)	0.80	0.64	0.71	2121
Accuracy			0.74	4141
Macro Avg	0.75	0.74	0.74	4141
Weighted Avg	0.75	0.74	0.73	4141

Ready for Real-Time Deployment

The migration from Sequence Processing (LSTM) to Statistical Vectorization (DNN) yielded a lightweight, exceptionally fast inference engine.

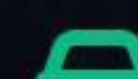
-  **Zero Latency:** Dense networks execute 10x faster than recurrent counterparts on edge devices.
-  **Noise Immunity:** Statistical smoothing naturally filters out blink artifacts and camera jitter.
-  **Next Steps:** Hardware integration via Raspberry Pi for live in-cabin testing.

Image Sources



<https://images.contentstack.io/v3/assets/blt1bd0ccbd0b7d1870/blt333f9d7c8a6c009d/68a45fcb7b073f3739e0c73c/falling-asleep-at-the-wheel.jpg>

Source: www.rac.co.uk



https://miro.medium.com/1*CIh_NdkVDHKwi4MnPpts1w.gif

Source: magdazelezik.medium.com



<https://images.presentationgo.com/2025/04/ai-neural-network-brain.jpg>

Source: www.presentationgo.com